

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12385588
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Staples; Tripp Capel et al.

Side entry sub

Abstract

A side entry sub has an elongated main body with a longitudinal bore and a side passage intersecting the longitudinal bore. A removable hub has a nose portion which threadably engages mating threads in the side passage, with a resilient seal on the nose portion providing a fluid seal with the main body. The threads are formed such that a flange on the hub advances into contact with the main body with a desired contact force as holes in the flange come into alignment with threaded holes in the main body. Threaded flange bolts through the flange holes engage the threaded holes in the main body. Locking washers are used on the flange bolts. The removable hub is structurally connected to the main body by the threaded connection and contact of the flange with the main body, and pressure and fluid-sealed by the resilient seal.

Inventors:	Staples; Tripp Capel (Youngsville, LA), Montet; Shaun Paul (Youngsville, LA), Hincapie; Juan Carlos (Broussard, LA), McKnight; Johnson M. (Youngsville, LA)
Applicant:	M&M International LLC (Cade, LA)
Family ID:	81754987
Assignee:	M&M INTERNATIONAL, LLC (Cade, LA)
Appl. No.:	18/036563
Filed (or PCT Filed):	November 24, 2021
PCT No.:	PCT/US2021/060688
PCT Pub. No.:	WO2022/115509
PCT Pub. Date:	June 02, 2022

Prior Publication Data

Document Identifier	Publication Date
----------------------------	-------------------------

Related U.S. Application Data

us-provisional-application US 63118839 20201127

Publication Classification

Int. Cl.: F16L41/14 (20060101); E21B17/042 (20060101)

U.S. Cl.:

CPC F16L41/14 (20130101); E21B17/0426 (20130101)

Field of Classification Search

CPC: F16L (41/04); F16L (41/045); F16L (41/06); F16L (41/065); F16L (41/086); F16L (41/14)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2713909	12/1954	Baker	N/A	N/A
3545542	12/1969	Scott	166/155	F16L 41/04
3931992	12/1975	Coel	285/271	F16L 41/04
4132243	12/1978	Kuus	15/104.062	F16L 41/04
4681162	12/1986	Boyd	N/A	N/A
4955406	12/1989	Antoniello	285/198	F16L 41/065
5054820	12/1990	Lesquir	47/52	F16L 41/04
5373866	12/1993	Whalen, II	222/5	F16L 41/06
8651174	12/2013	Barbee	166/70	F16L 41/04
8899254	12/2013	Weiler	81/53.2	F16L 41/06
9661807	12/2016	Turk	N/A	F16L 41/04
2004/0251018	12/2003	McLeod et al.	N/A	N/A
2007/0068679	12/2006	Robichaux et al.	N/A	N/A
2008/0271884	12/2007	Robichaux et al.	N/A	N/A
2015/0233514	12/2014	Kesler	285/208	F16L 41/06
2018/0259108	12/2017	Korus	N/A	F16L 41/04
2019/0017628	12/2018	Malasorti	N/A	F16L 41/04

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
19737353	12/1998	DE	F16L 41/04
19846088	12/1999	DE	F16L 41/06
102012109033	12/2013	DE	F16L 41/06
0611912	12/1993	EP	F16L 41/04
60921	12/1980	FI	F16L 41/04
08338586	12/1995	JP	F16L 41/06

10002477

12/1997

JP

F16L 41/06

20140046940

12/2013

KR

F16L 41/04

OTHER PUBLICATIONS

Commissioner for Patents, USPTO, as International Search Authority, International Search Report and Written Opinion for PCT/US2021/060688; Jul. 8, 2022; all pages; USPTO; Alexandria, Virginia, US. cited by applicant

Primary Examiner: Troutman; Matthew

Assistant Examiner: Kee; Fannie

Attorney, Agent or Firm: LAW OFFICE OF JESSE D. LAMBERT, LLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This United States non-provisional patent application claims priority to U.S. provisional patent application Ser. 63/118,839, filed Nov. 27, 2020, for all purposes. The disclosure of that provisional patent application is incorporated herein by reference, to the extent not inconsistent with this application.

BACKGROUND

Field of the Invention

(1) Various types of apparatus are used in connection with the drilling and servicing of oil and gas wells. At times, it is desired to have the ability to pump into a drillstring by a flowpath other than or in addition to the usual flowpath (directly down the drillstring bore), namely by a connection at the top of the drillstring. Such flowpath may be provided by a “side entry” sub, which is a tubular sub that comprises one or more fluid flowpaths into the bore of the drillstring through the wall of the sub, typically by a short tubular piece (which may be flanged) (which may be referred to as a “hub”) onto which a hose or similar flowline can be connected, hence the fluid flowing into the “side” of the drillstring.

(2) Prior art side entry subs present various issues, giving rise to a desire for an improved side entry sub that addresses these issues.

SUMMARY OF THE INVENTION

(3) The side entry sub of the present invention comprises an elongated main body, typically with a larger outer diameter central portion. The main body preferably comprises threaded connections or similar means on each end for connection within a drillstring. The main body has a longitudinal bore.

(4) A side passage, which comprises threads in a portion of its length, extends through the wall of the main body, connecting to the bore of the main body. The side entry element, which may be referred to as the “hub,” has a nose portion with mating threads to make up into the threads of the passage, and has a bore to permit fluid flow into the main body longitudinal bore. While various thread types are possible, preferably a straight or parallel (rather than tapered) thread form is employed. A resilient seal element, which may be an O-ring or similar seal element, is preferably disposed on the nose portion and provides a fluid and pressure seal between the hub and the main body.

(5) The hub additionally comprises a flange with a flange face which bears against the main body with a desired contact force, when the threaded connection between the main body and hub is fully made up. One or more, typically four, flange bolts further attach the hub flange to the main body.

Preferably, the flange bolts comprise a locking means to prevent back-out of the bolts. The locking means may comprise locking washers and wired bolts. Alternatively, and in a presently preferred embodiment, the locking means comprises NORD-LOCK® locking washers, providing an extra element of security to the connection.

(6) The outer end of the hub comprises a means for connecting a circulating hose or similar fluid line to the hub, for fluid passage through the hub into the bore of the main body. External hammer union threads are commonly used.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of the side entry sub embodying the principles of the present invention.

(2) FIG. 2 is a side view of the side entry sub of FIG. 1.

(3) FIG. 3 is an exploded, perspective view of the side entry sub.

(4) FIG. 4 is an exploded, side view of the side entry sub.

DESCRIPTION OF THE PRESENTLY PREFERRED EMBODIMENT(S)

(5) While various side entry subs can embody the principles of the present invention, with reference to the drawings some of the presently preferred embodiments can be described.

(6) FIGS. 1 and 2 are perspective and side assembled views, respectively, of side entry sub **10**, which may be referred to as “sub” **10**, embodying the principles of the present invention. Sub **10** comprises an elongated main body **20**, preferably with threaded connections **22** at either end, for makeup within a drillstring. As can be seen in FIG. 1, main body **20** comprises a longitudinal bore **24** for fluid passage through main body **20**. A side passage **26**, which may be better seen in FIG. 3, is positioned at a desired point along the length of main body **20** and provides a fluid flowpath through the wall of main body **20** into bore **24**. Passage **26** comprises internal threads **28** over a portion of its length. A removable hub **30** connects to main body **20**, as seen in FIGS. 1 and 2, and as described in more detail below.

(7) FIGS. 3 and 4 are exploded perspective and side views, respectively, of the various elements of side entry sub **10**. Hub **30** comprises a nose portion **31**, which comprises threads **32** to engage threads **28** in passage **26**. A seal groove **36** is positioned on nose portion **31**, into which a resilient seal element **37**, which may be an O-ring or similar resilient seal, is placed. Hub **30** further comprises a bore **34** and a flange **38**. The outermost end **39** of hub **30** preferably comprises coarse external threads **40** to enable connection of a hose or other fluid line thereto by a hammer union or similar connection means.

(8) Sub **10** is assembled by screwing hub **30** into side passage **26**. Resilient seal element **37** is first mounted in seal groove **36**. Preferably, threads **28** and **32** are “straight” or parallel threads, rather than tapered threads. Suitable thread forms may include ACME or similar type threads, which generally require low torque to fully engage the threads. Hub **30** is screwed into side passage **26** until flange **38** contacts main body **20**. During manufacture, the threads are cut (“timed” or “clocked”) so that bolt holes **54** in flange **38** align with the threaded bolt holes **56** in main body **20**, when flange **38** contacts main body **20** and suitable final torque is applied, yielding a desired contact force between flange **38** and main body **20**. Upon achieving this final rotational position with hub **30**, hub **30** is both pressure- and fluid-tight connected to main body **20** (by seal element **37** and threads **28** and **32**), and structurally connected to main body **20** via threads **28** and **32**, and contact of flange **38** with main body **20**.

(9) Flange bolts **50** can then be inserted through flange bolt holes **54** and made up into threaded holes **56** in main body **20**. While four flange bolts **50** are shown, it is understood that other numbers could be used. Preferably, locking washers **52** are used on flange bolts **50**, which may be NORD-

LOCK® or similar type locking washers.

(10) Sub **10** and its various components may be manufactured from high strength metal alloys as known in the art, and as needed to suit particular applications. Dimensions may be varied to meet particular applications. Methods of manufacture are known in the relevant art.

CONCLUSION

(11) While the preceding description contains many specificities, it is to be understood that same are presented only to describe some of the presently preferred embodiments of the invention, and not by way of limitation. Changes can be made to various aspects of the invention, without departing from the scope thereof.

(12) Therefore, the scope of the invention is to be determined not by the illustrative examples set forth above, but by the appended claims and their legal equivalents.

Claims

1. An apparatus, comprising: an elongated main body comprising a longitudinal bore therethrough, said longitudinal bore forming a wall in said main body, said main body further comprising a side passage extending through said wall and intersecting said longitudinal bore, said side passage comprising internal threads therein; a removable hub comprising a longitudinal bore therethrough, and further comprising: a nose portion comprising threads engageable with said internal threads in said side passage; a flange; and an outermost end comprising external threads; a resilient seal element sealing between said main body and said removable hub, when said removable hub is threadably engaged with said internal threads in said side passage of said main body; and wherein, when said removable hub is fully threadably engaged with said main body, said flange is in contact with said main body, and one or more fasteners are positioned through holes in said flange and into said main body, thereby preventing rotation of said removable hub.
2. The apparatus of claim 1, wherein said nose portion of said removable hub comprises a seal groove, and wherein said resilient seal element is disposed in said seal groove.
3. The apparatus of claim 2, wherein said one or more fasteners comprise threaded flange bolts extending through said holes in said flange and threadably engaging threaded holes in said main body.
4. The apparatus of claim 3, further comprising locking washers between said flange bolts and said flange.
5. The apparatus of claim 1, wherein said internal threads in said side passage and said threads of said nose portion of said removable hub comprise parallel thread forms.
6. The apparatus of claim 1, wherein said main body comprises threaded connections at either end.
7. A side entry sub, comprising: an elongated main body comprising a longitudinal bore therethrough, said longitudinal bore forming a wall in said main body, said main body further comprising a side passage extending through said wall and intersecting said longitudinal bore, said side passage comprising internal threads therein; a removable hub comprising a longitudinal bore therethrough, forming a flowpath through said removable hub and into said longitudinal bore of said main body when said removable hub is threadably engaged with said main body, wherein said removable hub further comprises a nose portion comprising threads engageable with said internal threads in said side passage and a seal groove, an outermost end comprising external threads, and a flange positioned between said nose portion and said outermost end; a resilient seal element positioned in said seal groove and sealing between said main body and said removable hub, when said removable hub is threadably engaged with said main body; and wherein, when said removable hub is fully threadably engaged with said main body, said flange is in contact with said main body, and one or more fasteners are positioned through holes in said flange and into said main body, thereby preventing rotation of said removable hub.
8. The side entry sub of claim 7, wherein said fasteners comprise threaded flange bolts extending

through said holes in said flange and threadably engaging threaded holes in said main body.

9. The side entry sub of claim 8, further comprising locking washers between said flange bolts and said flange.

10. The side entry sub of claim 8, wherein said threads in said side passage and said threads on said nose portion of said removable hub are formed so that said flange contacts said main body with a desired contact force when said holes in said flange are aligned with said threaded holes in said main body.

11. The side entry sub of claim 10, wherein said internal threads in said side passage and said threads of said nose portion of said removable hub comprise parallel thread forms.

12. The side entry sub of claim 11, wherein said external threads on said outermost end of said removable hub comprise hammer union threads.

13. The side entry sub of claim 12, wherein said main body comprises threaded connections at either end.
